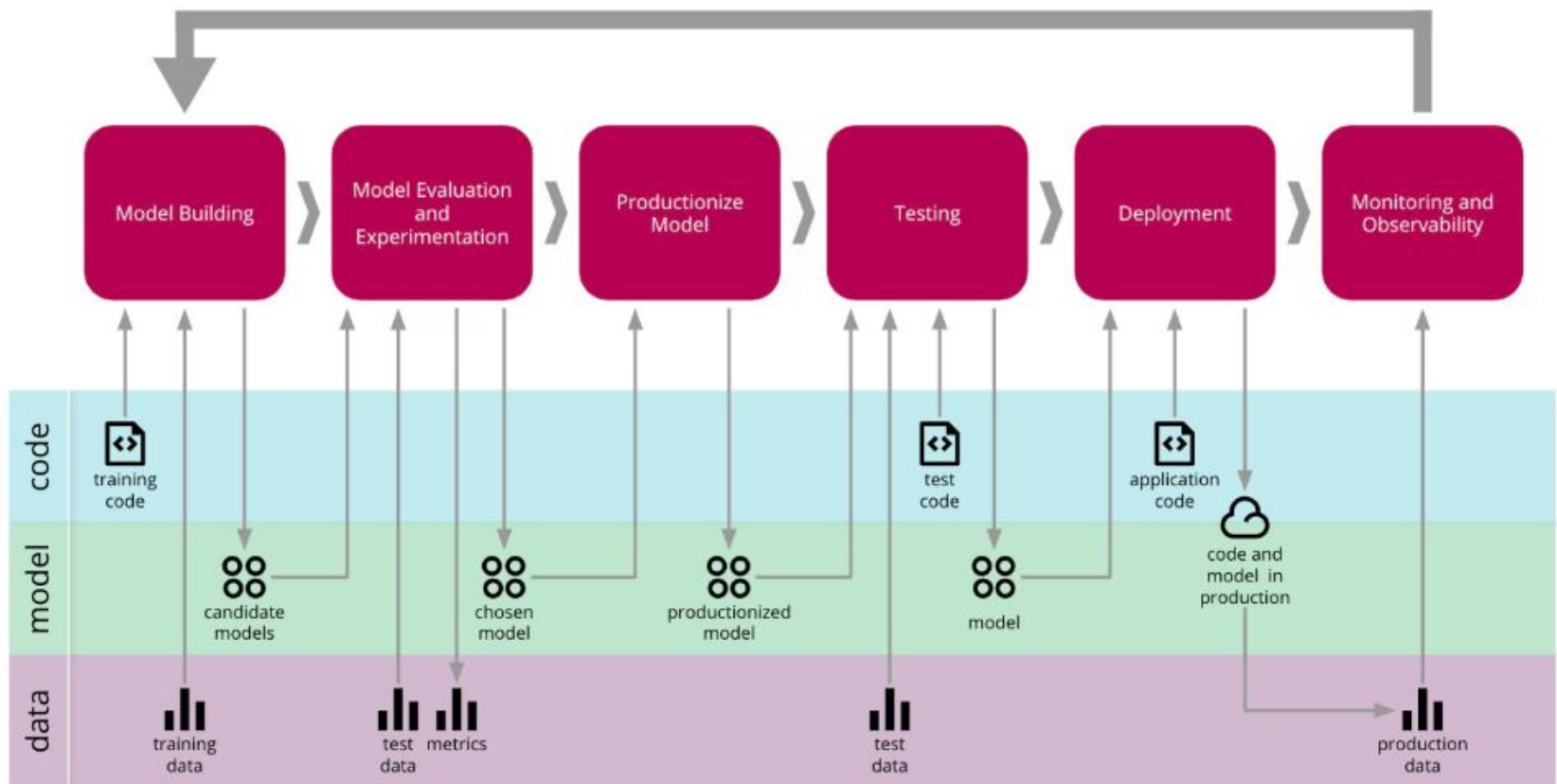


Workflows e Rastreamento

Luciano Barbosa

Workflow de Machine Learning

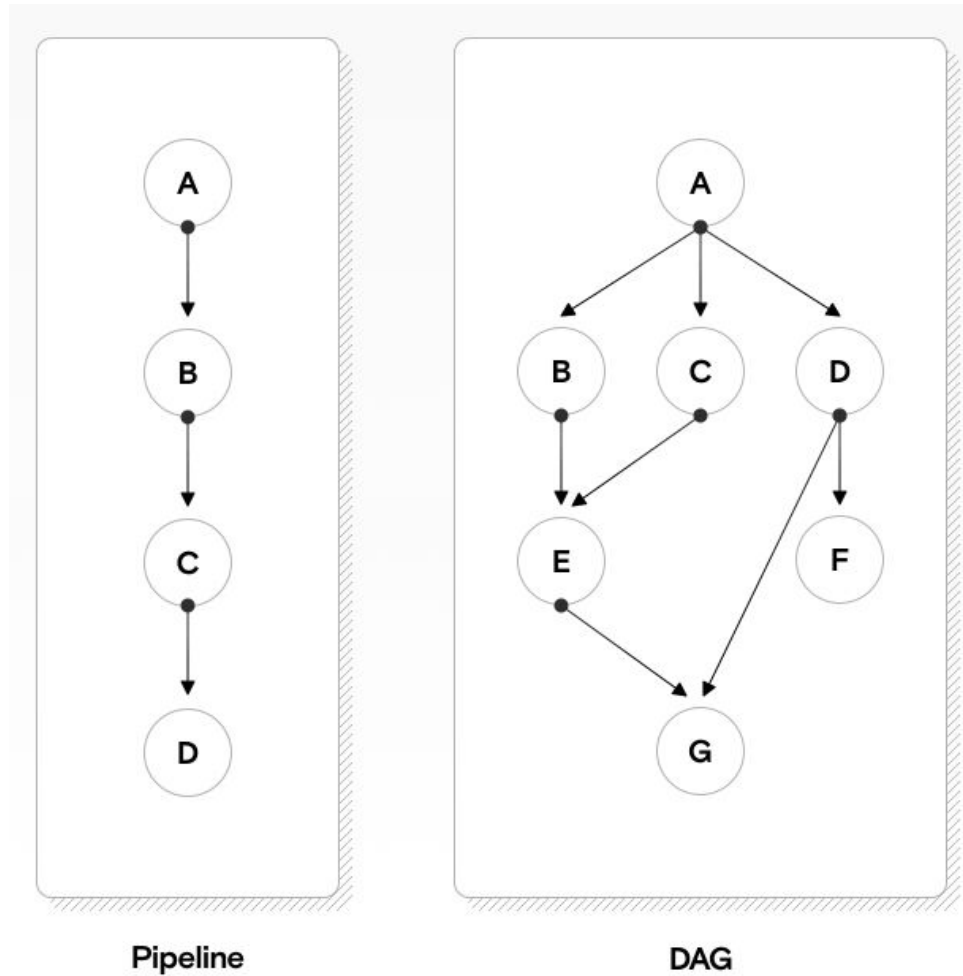


ETL



Source: [Vineet Goel's "Why Robinhood uses Airflow?" Medium Post](#)

Directed Acyclic Graph - DAG



Airflow: Operadores

- Sensors: inicia a tarefa
 - Agendado no tempo
 - Dados de entrada disponíveis
- Operators: executa a tarefa
 - Python, docker, mysql, s3, email, http, spark
- Transfers: transfere dados de um lugar para outro
 - google_api_to_s3_transfer

Airflow

 **Airflow**

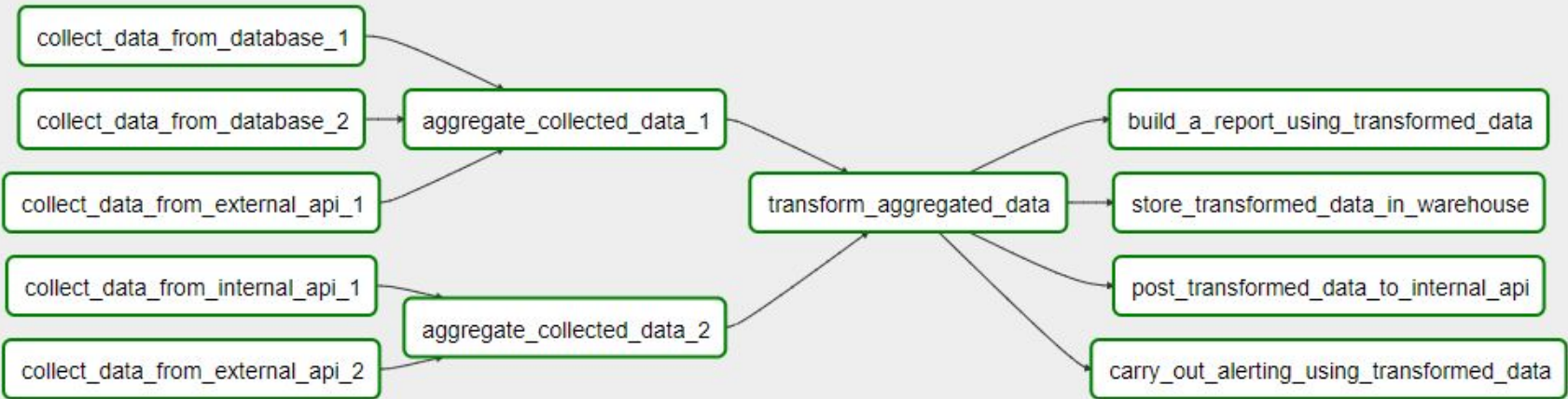
[DAGs](#) [Data Profiling](#) [Browse](#) [Admin](#) [Docs](#) [About](#)

On **DAG: example_dag**

[Graph View](#) [Tree View](#) [Task Duration](#) [Task Tries](#) [Landing Times](#) [Gantt](#) [Details](#) [Code](#) [Refresh](#) [Delete](#)

running Base date: Number of runs: Run: Layout:

PythonProgramOperator



```
graph LR; C1[collect_data_from_database_1] --> A1[aggregate_collected_data_1]; C2[collect_data_from_database_2] --> A1; C3[collect_data_from_external_api_1] --> A1; C4[collect_data_from_internal_api_1] --> A2[aggregate_collected_data_2]; C5[collect_data_from_external_api_2] --> A2; A1 --> T[transform_aggregated_data]; A2 --> T; T --> B[build_a_report_using_transformed_data]; T --> S[store_transformed_data_in_warehouse]; T --> P[post_transformed_data_to_internal_api]; T --> A3[carry_out_alerting_using_transformed_data]
```

Airflow

DAGs

All 26		Active 10	Paused 16	Filter DAGs by tag		Search DAGs	
DAG	Owner	Runs	Schedule	Last Run	Recent Tasks	Actions	Links
☒ example_bash_operator example example2	airflow	2	0 0 ***	2020-10-26, 21:08:11	6	▶ ↺ 🗑️ ...	
☒ example_branch_dop_operator_v3 example	airflow		*/1 * * * *			▶ ↺ 🗑️ ...	
☐ example_branch_operator example example2	airflow	1	@daily	2020-10-23, 14:09:17	11	▶ ↺ 🗑️ ...	
☒ example_complex example example2 example3	airflow	1 1	None	2020-10-26, 21:08:04	37	▶ ↺ 🗑️ ...	
☒ example_external_task_marker_child	airflow	1	None	2020-10-26, 21:07:33	2	▶ ↺ 🗑️ ...	
☒ example_external_task_marker_parent	airflow	1	None	2020-10-26, 21:08:34	1	▶ ↺ 🗑️ ...	
☒ example_kubernetes_executor example example2	airflow		None			▶ ↺ 🗑️ ...	
☒ example_kubernetes_executor_config example3	airflow	1	None	2020-10-26, 21:07:40	5	▶ ↺ 🗑️ ...	
☒ example_nested_branch_dag example	airflow	1	@daily	2020-10-26, 21:07:37	9	▶ ↺ 🗑️ ...	
☐ example_passing_params_via_test_command example	airflow		*/1 * * * *			▶ ↺ 🗑️ ...	

Airflow

 DAG: example_complex

schedule: None

 Tree View

 Graph View

 Task Duration

 Task Tries

 Landing Times

 Gantt

 Details

 Code



2020-10-01T21:00:14-C

Runs

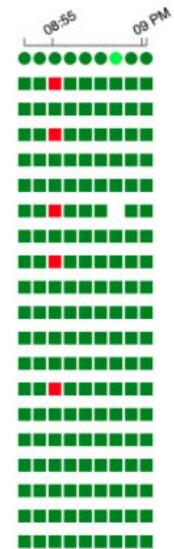
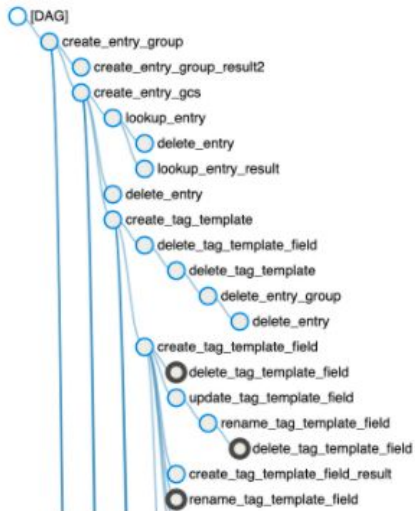
25



Update

 BashOperator  PythonOperator

 scheduled  skipped  upstream_failed  up_for_reschedule  up_for_retry  failed  success  running  queued  no_status



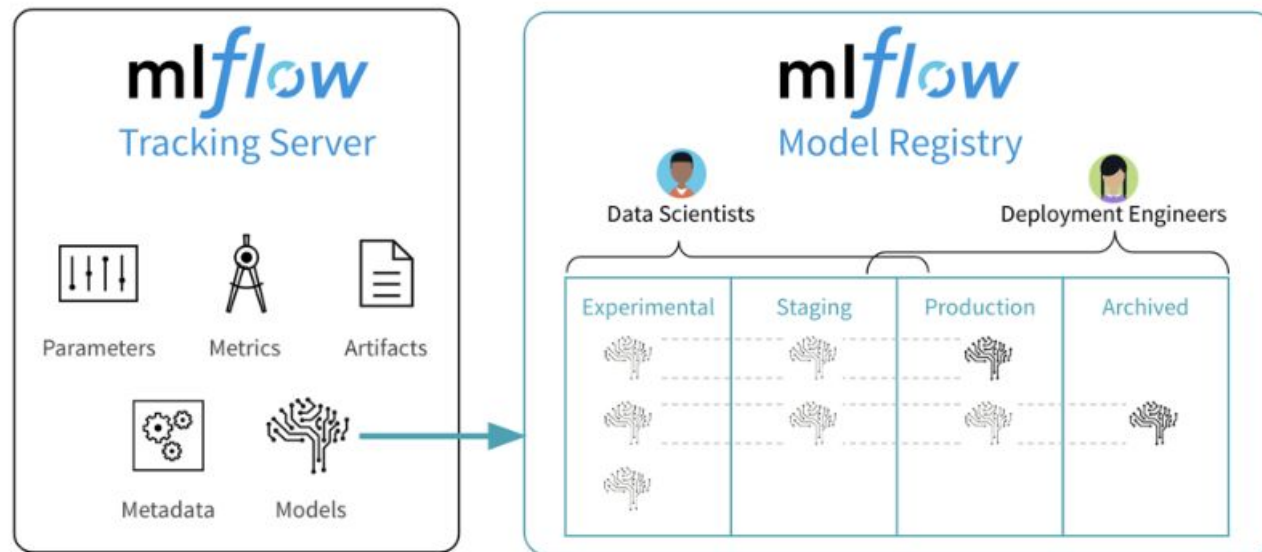
Rastreamento de Resultados

Rastreando Experimentos com Planilhas

11	Dataset	Split (train/dev/test)	0.7/0.2/0.1	0.7/0.2/0.1	0.7/0.2/0.1	0.7/0.15/0.15	0.7/0.15/0.15
12		Class ratio (train/dev/test)	0.42/0.42/0.42	0.42/0.42/0.42	0.42/0.42/0.42	/0.3/0.3	/0.3/0.3
13		train/dev/test size	4871/1392/696	4871/1392/696	5315/1518/760	5315/1139/1139	5315/1139/1139
14	Training hyperparameters	Learning rate	1.00E-05	1.00E-05	1.00E-05	1.00E-05	1.00E-05
15		epoch	3	2	1	5	6
16		batch size	32	32	32	32	32
17	Results	accuracy	0.88304595	0.8650862069	0.8687747	0.86997364	0.65
18		f1	0.82495437	0.8108753316	0.82383946	0.81827954	0.44
19		precision	0.878865	0.7848381601	0.8407407	0.8556561	0.56
20		recall	0.7780239	0.8389705882	0.8076923	0.78442625	0.36
21		tp	1398	1402	1460	1334	1130
22		tn	1692	1663	1707	1543	1504
23		fp	1113	1142	1161	1108	1148
24		fn	1189	1185	1190	1154	1357
25		loss	0.59637538	0.594134	0.594134	0.6037084	0.594134
26	Test results	accuracy	0.90747	0.90747	0.88026	0.88314	0.75847
27		f1	0.85636	0.85636	0.83108	0.83469	0.5915
28		precision	0.90934	0.90934	0.86689	0.87027	0.77626
29		recall	0.8099	0.8099	0.79846	0.80226	0.48604
30							

MLflow

- Ferramenta para rastreamento automático de resultados



Comparando Modelos

	Date	User	Source	Version	Parameters		Metrics		
					alpha	l1_ratio	mae	r2	rmse
☐	2018-06-04 23:00:10	mlflow	train.py	05e956	1	1	0.649	0.04	0.862
☐	2018-06-04 23:00:10	mlflow	train.py	05e956	1	0.5	0.648	0.046	0.859
☐	2018-06-04 23:00:10	mlflow	train.py	05e956	1	0.2	0.628	0.125	0.823
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	1	0	0.619	0.176	0.799
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	1	0.648	0.046	0.859
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	0.5	0.628	0.127	0.822
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	0.2	0.621	0.171	0.801
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	0	0.615	0.199	0.787
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	0	1	0.578	0.288	0.742
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	0	0.5	0.578	0.288	0.742
☐	2018-06-04 23:00:09	mlflow	train.py	05e956	0	0.2	0.578	0.288	0.742
☐	2018-06-04 23:00:08	mlflow	train.py	05e956	0	0	0.578	0.288	0.742

Tela de Rastreamento

Run 7c1a0d5c42844dcdb8f5191146925174

Experiment Name: Default

Start Time: 2018-06-04 23:47:22

Source: train.py

Git Commit: 3aa48cffe58b8d9d69f5

User: mlflow

Duration: 145ms

▼ Parameters

Name	Value
alpha	0
l1_ratio	0

▼ Metrics

Name	Value
mae	0.578
r2	0.288
rmse	0.742

► Tags

▼ Artifacts

▼

model

MLmodel

model.pkl

Full Path:/Users/mlflow/mlflow-prototype/mlruns/0/7c1a0d5c42844dcdb8f5191146925174/artifacts/model/MLmodel

Size: 259B

artifact_path: model

flavors:

python_function:

data: model.pkl

loader_module: mlflow.sklearn

sklearn:

pickled_model: model.pkl

sklearn_version: 0.19.1

run_id: 7c1a0d5c42844dcdb8f5191146925174

utc_time_created: '2018-06-05 06:47:22.757025'